FLAG REFINEMENT OF THE HYPERSIMPLEX H-STAR FORMULA AND YOUNG-MODULE INTERPRETATION

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Editorial note. Complete original mathematical paper91–873, all nonadjacent supporting proofs and unchanged main.bib retained. Authors retain responsibility for mathematics; no proof repair, added assumptions, independent author review or certification.

Editorial note. Original flag-h and Young-module main statements form one combined refinement object. Their coprimality and cyclic-composition conventions remain exact; intermediate generating-function/contingency identities and direct consequences are full proof support, not extra counts.

Editorial note. Focused public neighbor1001 concerns a different integral matroid Chow-ring tree-expansion theorem, not this hypersimplex alcove flag enumeration. No proof re-audit performed.

Allowed standard prerequisites (editorial).

- Hypersimplex alcove triangulations, decorated ordered partitions, cyclic flags and symmetric functions. Original k/n, cyclic composition, balanced coloring, flag h/f, Young module and gcd(k,n)=1 hypotheses retained; known Early-Kim theorem and alcove description remain explicitly cited. All source-local symmetric-function/bijection proofs supplied.. Complete original paper and main.bib retained; no independent verification, added hypotheses or proof reconstruction.

1. Introduction

The hypersimplex $\Delta_{k,n} = \{x \in [0,1]^n : \sum_i x_i = k\}$ is a $(n-1)$ -dimensional polytope with $\binom{n}{k}$ vertices. The *Ehrhart polynomial* $\mathrm{Ehr}(P,t)$ of a lattice polytope P is defined such that

$$\mathrm{Ehr}(P,t) = |tP \cap \mathbb{Z}^n|, \qquad \text{for } t \in \mathbb{N}.$$

The h^**-polynomial* of P is defined by the generating function

$$\sum_{t\geq 0} \mathrm{Ehr}(P,t) z^t = \frac{h^*(P,z)}{(1-z)^{\dim(P)+1}}.$$

Early conjectured a combinatorial formula for the h^*-polynomial of $\Delta_{k,n}$ in terms of *hypersimplicial decorated ordered set partitions*, which was later proved by Kim [Kim20].

Definition 1.1. Let $(S_1, \ldots, S_p)$ be an ordered set partition of $[n]$ and $(s_1, \ldots, s_p)$ be a composition of k. There is an equivalence relation on these pairs, defined by

$$((S_1)_{s_1}, (S_2)_{s_2}, \ldots, (S_p)_{s_p}) \sim ((S_2)_{s_2}, \ldots, (S_p)_{s_p}, (S_1)_{s_1}).$$

A *decorated ordered set partition*

$$\mathsf{S} = [((S_1)_{s_1}, \ldots, (S_p)_{s_p})] \tag{1}$$

is an equivalence class of such pairs. A decorated ordered set partition is *hypersimplicial* if $1 \leq s_i \leq |S_i| - 1$ for all i. We denote the set of hypersimplicial decorated ordered set partitions of type (k,n) by $\mathrm{hDOSP}(k,n)$. We call each S_i a block, place the blocks on a circle clockwise, and regard s_i as the clockwise distance between adjacent blocks S_i and S_{i+1}. The *winding number* of S is defined as the number of times we wrap around the circle as we trace $1 \to 2 \to \cdots \to n \to 1$ clockwise.

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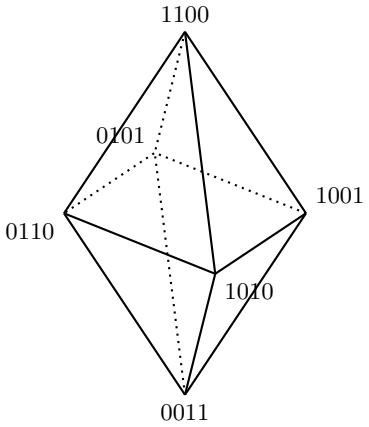


FIGURE 2. The hypersimplex $\Delta_{2,4}$.

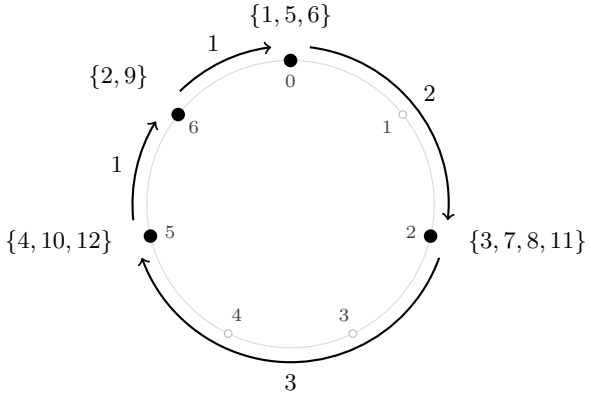


FIGURE 1. The decorated ordered set partition $(\{1,5,6\}_2, \{3,7,8,11\}_3, \{4,10,12\}_1, \{2,9\}_1)$ of type $(7,12)$ with winding number 5.

Remark 1.2. Let $(S_1, \ldots, S_p)$ be an ordered set partition of $[n]$, and $(s_1, \ldots, s_p)$ be a composition of k. For a set S, let $x_S = \sum_{i \in S} x_i$. Derksen and Fink [DF10] showed that the cones determined by the inequalities

$$\begin{aligned} x_{S_1} &\leq s_1 \\ x_{S_1 \cup S_2} &\leq s_1 + s_2 \\ &\vdots \\ x_{[n]} &\leq k. \end{aligned} \tag{2}$$

form a basis of the valuative ring of matroids on the ground set $[n]$ of rank k.

Theorem 1.3 ([Ear17, Kim20])**.** *The h^* -polynomial of the hypersimplex $\Delta_{k,n}$ is given by*

$$h^*(\Delta_{k,n}, z) = \sum_{\mathsf{S} \in \mathrm{hDOSP}(k,n)} z^{\mathrm{wind}(\mathsf{S})}.$$

Example 1.4. The hypersimplex $\Delta_{2,4}$ is the octahedron shown in Figure 2. The hypersimplicial decorated ordered set partitions of type $(2,4)$ are shown in Figure 3, and we have

$$h^*(\Delta_{2,4}, z) = 1 + 2z + z^2.$$

1.1. **Balanced complexes and flag numbers.** There is a *unimodular triangulation* of the hypersimplex $\Delta_{k,n}$ by *alcoves*, studied by Postnikov and Lam in [LP07]. The *alcove triangulation complex* of $\Delta_{k,n}$, denoted $\mathcal{T}_{k,n}$, is the simplicial complex with vertex set $\binom{[n]}{k}$, and maximal faces called *alcoves* (See Section 2). Since $\mathcal{T}_{k,n}$ is unimodular, Stanley's result in [Sta80] implies that

$$h^*_d(\Delta_{k,n}) = h_d(\mathcal{T}_{k,n}).$$

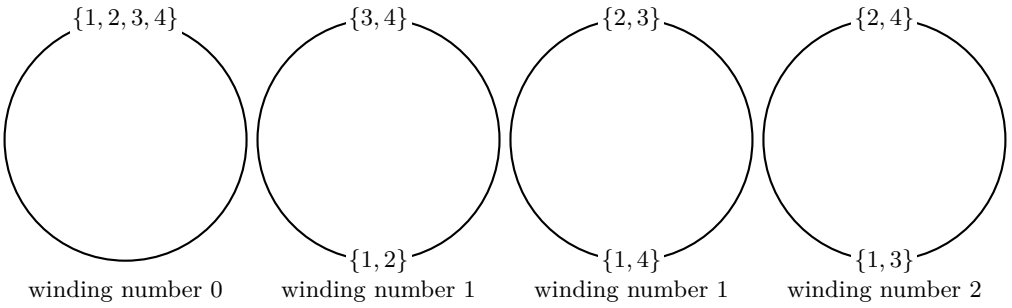


FIGURE 3. hDOSP(2, 4).

Our main result gives a refinement of Theorem 1.3 by flag numbers introduced by Stanley in the study of balanced complexes [Sta79].

Let Δ be a simplicial complex on a vertex set V. A face with $i+1$ elements is an *i-face* and has $\dim F = i$; if $\delta = \max\{\dim F : F \in \Delta\}$ then Δ is a *δ-complex* and $\dim \Delta = \delta$, and Δ is *pure* if every maximal face is a δ-face [Sta79, §1].

Definition 1.5 ([Sta79, §2])**.** A *balanced complex* of type $(a_1, \ldots, a_m)$ is a pair (Δ, π) such that Δ is a pure δ -complex on a vertex set V and $\pi = (V_1, \ldots, V_m)$ is an ordered set partition of V such that for every maximal face $F \in \Delta$ and every $i \in [m]$, we have $|F \cap V_i| = a_i$. Hence $a_1 + \cdots + a_m = \delta + 1$. A balanced complex of type $(1, 1, \ldots, 1)$ is *completely balanced*.

Definition 1.6 ([Sta79, §3])**.** Let (Δ, π) be completely balanced with $\pi = (V_1, \ldots, V_m)$.The *color* of a vertex $x \in V_i$ is $\mathrm{color}(x) = i$, and the *type* of a face $F \in \Delta$ is

$$\mathrm{type}(F) = \{\mathrm{color}(x) : x \in F\} \subseteq [m]. \tag{3}$$

For $S \subseteq [m]$ the *flag f-number* and the *flag h-number* of Δ are

$$f_S(\Delta) = \#\{F \in \Delta : \mathrm{type}(F) = S\}, \qquad h_S(\Delta) = \sum_{T \subseteq S} (-1)^{|S \setminus T|} f_T(\Delta). \tag{4}$$

Lemma 1.7 (Eq.6,[Sta79])**.** *If Δ is a completely balanced complex, then*

$$h_d(\Delta) = \sum_{|S|=d} h_S(\Delta). \tag{5}$$

1.2. **Cyclic compositions and cyclic words.** A *(resp. weak) composition* of n with p parts is a vector $(\alpha_1, \ldots, \alpha_p)$ of (resp. nonnegative) positive integers such that $\sum_{i=0}^{p-1} \alpha_i = n$, denoted $\alpha \models n$. There is an equivalence relation on (weak) compositions of n with p parts, induced by

$$(\alpha_1, \ldots, \alpha_p) \sim \alpha^{(i)} := (\alpha_i, \ldots, \alpha_p, \alpha_1 \ldots, \alpha_{i-1}).$$

The equivalence class under this relation is called a *cyclic composition*, denoted $[\alpha]$.

Definition 1.8. For each subset $S \subseteq [n]$ and for each $i \in S$, we associate a cyclic composition, denoted $\mathrm{comp}^{(i)}(S)$. We draw the points in S clockwise on a circle, so that they cut the circle into $|S|$ arcs. Let $\mathrm{comp}^{(i)}(S)_j$ be the arc length of the j-th arc clockwise after i. These arc lengths form a cyclic composition of n. The set $\{\mathrm{comp}^{(i)}(S) : i \in S\}$ forms an equivalence class of compositions, which we denote by $\mathrm{ccomp}(S)$. Conventionally, let $\mathrm{comp}(S) := \mathrm{comp}^{\min S}(S)$.

Let $w : \mathbb{Z}/n\mathbb{Z} \to [k]$ be a word of length n and alphabet $[k]$ for some positive integer k. Following [PSBTW25], we define the *cyclic descents* of w as

$$\mathrm{cDes}(w) := \{j \in \mathbb{Z}/n\mathbb{Z} : w(j) > w(j+1)\}, \qquad \mathrm{cdes}(w) = |\,\mathrm{cDes}(w)| \tag{6}$$

The *content* of w is the *weak* composition

$$\mathrm{content}(w) := (|w^{-1}(1)|, \ldots, |w^{-1}(k)|). \tag{7}$$

Definition 1.9. Let α be a weak composition with k parts. Viewing $\alpha : \mathbb{Z}/k\mathbb{Z} \to \mathbb{Z}_{\geq 0}$, we say that $\alpha \models n$ is *hypersimplicial* if every positive entry satisfies $\alpha_i > 1$ and is followed by at most $\alpha_i - 1$ zeros on the circle. A cyclic word is *hypersimplicial* if it has hypersimplicial content.

Let w be a cyclic word $\mathbb{Z}/n\mathbb{Z} \to [k]$. For $p \in \mathbb{Z}/k\mathbb{Z}$, define the *$p$-shift* of w as the cyclic word

$$w^{(p)}(i) = w(i) - p \mod k. \tag{8}$$

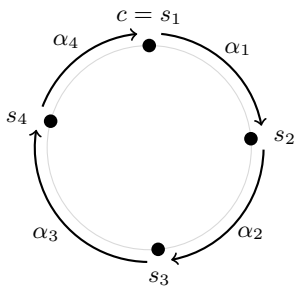


FIGURE 4. Reading the gaps of $S = \{s_1 < s_2 < s_3 < s_4\} \subseteq \mathbb{Z}/n\mathbb{Z}$: walking clockwise from $c = s_1$, the points of S appear in the order s_1, s_2, s_3, s_4, and α_i is the number of steps of the i-th arc, giving $\mathrm{comp}^c(S) = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$.

Definition 1.10. The *reading word* of an ordered set partition$(S_1, \ldots, S_p)$ is the cyclic word w defined by

$$w(i) = j \text{ if } i \in S_j. \tag{9}$$

For a decorated ordered set partition in the form (1), choose the representative such that $n \in S_1$, and define w_{S} as the reading word of this representative. Let $\mathrm{cDes}(\mathsf{S}) := \mathrm{cDes}(w_{\mathsf{S}})$.

1.3. **Main results.** Our first main result is a refinement of Early–Kim's theorem by giving a combinatorial formula for Stanley's flag h-numbers of the alcove triangulation complex.

Theorem 1.11. *Let $\mathcal{T}_{k,n}$ be the triangulation complex of $\Delta_{k,n}$. Then, for each cyclic composition α, we have*

$$\sum_{S:\mathrm{ccomp}(S)=[\alpha]} h_S(\mathcal{T}_{k,n}) = \left|\{\mathsf{S} \in \mathrm{hDOSP}(k,n) : \mathrm{ccomp}(\mathrm{cDes}(\mathsf{S})) = [\alpha]\}\right|. \tag{10}$$

Our second result gives a representation theoretic interpretation of the flag f-numbers. Let $\alpha = (\alpha_1, \ldots, \alpha_r)$ be a weak composition of n. The *Young subgroup* is $S_\alpha \cong S_{\alpha_1} \times \cdots \times S_{\alpha_r}$, and the *Young permutation module* of α is $M^\alpha \cong \mathrm{Ind}_{S_\alpha}^{S_n} \mathbf{1}$, the S_n -module on the ordered set partitions $(B_1, \ldots, B_r)$ of $[n]$ with $|B_i| = \alpha_i$. The Frobenius character of M^α is $\mathrm{ch}\, M^\alpha = h_\alpha$, the complete homogeneous symmetric function. Let

$$\mathcal{M}_{k,n} = \bigoplus_{\lambda \text{ hypersimplicial}} M^\lambda. \tag{11}$$

Theorem 1.12. *Let $\gcd(k,n) = 1$ and let S be a nonempty subset of $[n]$ with $\mathrm{ccomp}(S) = [\alpha]$. Then*

$$k\, f_S(\mathcal{T}_{k,n}) = \dim \mathrm{Hom}_{S_n}(M^\alpha, \mathcal{M}_{k,n}) \tag{12}$$

Organization. In Section 2, we introduce the alcoved triangulation of the hypersimplices. In Section 3, we prove symmetric function identities for the generating functions of the statistics we are interested in. In Section 4, we prove our main result.

Acknowledgements. We thank Nick Early, Elisabeth Bullock, Donghyun Kim, Lauren Williams, and Sylvie Corteel for helpful discussion.

2. The alcove triangulation

We introduce the alcove triangulation of the hypersimplex. Lam and Postnikov [LP07] showed that the maximal faces of $\mathcal{T}_{k,n}$ are parametrized by minimal *circuits*. Parisi, Sherman-Bennett, Tessler, and Williams [PSBTW25] showed that the circuits can be described as the cyclic descents set of permutations.

Definition 2.1 ([LP07])**.** A sequence $(I_1, \ldots, I_n)$ of k -subsets of $[n]$ is a *circuit* if for each $i \in \mathbb{Z}/n\mathbb{Z}$, there exists $j \in \mathbb{Z}/n\mathbb{Z}$ such that $I_{i+1} = I_i \setminus \{j\} \cup \{j+1\}$.

Theorem 2.2 ([LP07, PSBTW25])**.** *If $(I_0, \ldots, I_{n-1})$ is a maximal face in $\mathcal{T}_{k,n}$, then there exists a permutation $w \in S_n$ with $\mathrm{cdes}(w) = k$ such that for any $p \in \mathbb{Z}/n\mathbb{Z}$, we have*

$$I_p = I_{p-1} \setminus \{w^{-1}(p) - 1\} \cup \{w^{-1}(p)\} \tag{13}$$

for all $p \in \mathbb{Z}/n\mathbb{Z}$. Moreover, we may take $w(n) = 1$ and

$$I_p = \mathrm{cDes}\left(w^{(p)}\right). \tag{14}$$

w	$(I_0, \ldots, I_3)$
4312	(12,13,14,24)
4213	(12,13,23,24)
4231	(13,14,24,34)
4132	(13,23,24,34)

TABLE 1. The permutations and vertex sets indexing simplices in the hypersimplex $\Delta_{2,4}$.

Example 2.3. In Figure 2, the four simplices triangulating the hypersimplex $\Delta_{2,4}$ are shown in Table 1. Take the second row in that table where $w^{(0)} = 4213$ with $\mathrm{cDes}(w^{(0)}) = \{1, 2\}$. We have

$$w^{(1)} = 3142, \qquad \mathrm{cDes}(w^{(1)}) = \{1, 3\},$$
$$w^{(2)} = 2431, \qquad \mathrm{cDes}(w^{(2)}) = \{2, 3\},$$
$$w^{(3)} = 1324, \qquad \mathrm{cDes}(w^{(3)}) = \{2, 4\}.$$

Theorem 2.4. *The alcove triangulation complex $\mathcal{T}_{k,n}$ of $\Delta_{k,n}$ is completely balanced with respect to the coloring* $\mathrm{color}(I) :\equiv \sum_{i\in I} i \bmod n$.

Proof. Suppose F is a maximal face with circuits $(I_0, \ldots, I_{n-1})$, then

$$\mathrm{color}(I_p) \equiv \mathrm{color}(I_{p-1}) + 1 \bmod n, \qquad \text{for any } p \in \mathbb{Z}/n\mathbb{Z},$$

so all the vertices of F are of different colors, and each color appears only once. Any lower dimensional face is contained in a maximal face, and this property is inherited. □

Let $\mathbb{Z}/n\mathbb{Z}$ acts on each $S \subseteq [n]$ by

$$S + j := \{s + j \bmod n \colon s \in S\}, \qquad \text{for each } j \in \mathbb{Z}/n\mathbb{Z}. \tag{15}$$

Proposition 2.5. *Let $d = \gcd(k, n)$. For every $S \subseteq \mathbb{Z}/n\mathbb{Z}$, the flag numbers satisfy $f_S(\mathcal{T}_{k,n}) = f_{S+k}(\mathcal{T}_{k,n})$ and $h_S(\mathcal{T}_{k,n}) = h_{S+k}(\mathcal{T}_{k,n})$.*

Proof. Consider the rotation of the n-coordinates

$$\rho : \mathbb{R}^n \to \mathbb{R}^n, (x_1, \ldots, x_n) \mapsto (x_2, \ldots, x_n, x_1)$$

which preserves the alcove triangulation. Then $\mathrm{type}(\rho F) = \mathrm{type}(F) + k$ and $f_S(\mathcal{T}_{k,n}) = f_{S+k}(\mathcal{T}_{k,n})$. Applying Möbius transformation, we obtain the same equality for h_S. □

As k generates a subgroup of $\mathbb{Z}/n\mathbb{Z}$ with index $\gcd(k, n)$, we have the following corollary.

Corollary 2.6. *The functions $j \mapsto f_{S+j}(\mathcal{T}_{k,n})$ and $j \mapsto h_{S+j}(\mathcal{T}_{k,n})$ only depend on $j \bmod d$ and are constant when $\gcd(k, n) = 1$.*

For any cyclic word w on $\mathbb{Z}/n\mathbb{Z}$, let

$$\mathrm{cEq}(w) = \{j \in \mathbb{Z}/n\mathbb{Z} : w(j) = w(j + 1)\}, \qquad \mathrm{ceq}(w) = |\,\mathrm{cEq}(w)|, \tag{16}$$

denote its set of *cyclic equalities*.

Lemma 2.7. *For any face $F \in \mathcal{T}_{k,n}$ of dimension $p - 1$,*

1. *there exists an ordering $F = (I_0, \ldots, I_{p-1})$ of its vertices such that I_0 is lexicographically minimal, denoted* $\mathrm{base}(F) := I_0$. *Moreover, let $\alpha_i = e_i - e_{i-1}$ for $i \in \{2, \ldots, n\}$, then for any $p \geq 1$, we let $I_p - I_{p-1} = \alpha_{A_i} := \sum_{j\in A_i} \alpha_j$, for some subset $A_i \subseteq [n]$.*
2. *The A_i's are mutually disjoint. Let $A_p := [n] \setminus \left(\bigcup_{i=1}^{p-1} A_i\right)$.*
3. *Let $w_F : [n] \to [p]$ be the word defined by*

$$w_F(i) = \min\{t : i \in A_1 \cup \cdots \cup A_t\}, \tag{17}$$

then

$$\mathrm{cDes}(w_F) \subseteq \mathrm{base}(F) \subseteq \mathrm{cDes}(w_F) \cup \mathrm{cEq}(w_F). \tag{18}$$

Conversely, for any word $w : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ with $\mathrm{cdes}(w) \leq k$ and any k -subset I in $[n]$ such that

$$\mathrm{cDes}(w) \subseteq I \subseteq \mathrm{cDes}(w) \cup \mathrm{cEq}(w), \tag{19}$$

there is a face $F = (I = I_1, \ldots, I_p)$ for which $w = w_F$.

(4) *Fix some word* $w : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$. *The number of* $I \in \binom{[n]}{k}$ *satisfying* (19) *is*

$$\binom{\mathrm{ceq}(w)}{k - \mathrm{cdes}(w)}.$$

Proof.

(1) Suppose that $F = (I_0, \ldots, I_{n-1})$ is a maximal face, and w is a permutation taken from Theorem 2.2 with $w(n) = 1$. By (13), we see that I_p is lexicographically greater than I_{p-1} for $p \geq 1$. If F is a face of lower dimension, then we may embed it into a maximal face, and the conclusion follows from the case for maximal faces.
(2) For a maximal face F with $w \in S_n$ as in Theorem 2.2, we have $A_i = \{w^{-1}(i)\}$. They are mutually disjoint because w is a permutation. For lower dimensional faces, we embed them into maximal faces to conclude.
(3) Suppose that F is a maximal face, and $w \in S_n$ is as in Theorem 2.2. Moreover, by (1), we may assume that $w(n) = 1$. Then $w_F = w$ and $\mathrm{base}(F) = \mathrm{cDes}(w)$ by (14). If F is lower dimensional, we may embed F into a maximal face F' with a permutation w with $w(n) = 1$ as in Theorem 2.2 and (1). Consequently, $\mathrm{base}(F') = \mathrm{cDes}(w)$. As $F \subseteq F'$, the tuple $(A_1, \ldots, A_p)$ is an ordered set partition of which w is a refinement, and w_F is the reading word (9) of $(A_1, \ldots, A_p)$. The set $\mathrm{cDes}(w_F)$ consists of indices i that appear in a later block than $i+1$, and the set $\mathrm{cEq}(w_F)$ consists of indices i that appear in the same block as $i+1$. As w is a refinement of $(A_1, \ldots, A_p)$, the conclusion follows. Conversely, given the word $w : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$, let A_i be the preimage of i under w. We can form $(I_1, \ldots, I_p)$ inductively by setting I_1 to I, and use $I_p = I_{p-1} + \alpha_{A_i}$.
(4) We may assume that $\mathrm{cdes}(w) \leq k$. The condition on I means that $I \setminus \mathrm{cDes}(w)$ can be any subset of $\mathrm{cEq}(w)$ with cardinality $k - \mathrm{cdes}(w)$, hence the conclusion.

□

Example 2.8. Choose $F = (I_0, I_1)$ such that $I_0 = \{1, 2\}, I_1 = \{2, 4\}$, and embed F into the face in Example 2.3. We have $I_1 - I_0 = (e_3 - e_2) + (e_4 - e_3) = \alpha_3 + \alpha_4$ so $A_1 = \{3, 4\}$ and $A_2 = \{1, 2\}$. The reading word of this ordered set partition is $w_F = 2211$. We have $\mathrm{cDes}(w_F) = \{2\}$ and $\mathrm{cEq}(w_F) = \{1, 3\}$.

For a face $F \in \mathcal{T}_{k,n}$, let

$$\mathrm{comp}(F) := \alpha(\mathrm{type}(F)^{\mathrm{color}(\mathrm{base}(F))}). \tag{20}$$

For a composition α, let $W(\alpha)$ be the set of cyclic words of content α.

Proposition 2.9. *Let* $\alpha \models n$ *be a composition with* p *parts. The number of nonempty faces* $F \in \mathcal{T}_{k,n}$ *with* $\mathrm{comp}(F) = \alpha$ *is equal to*

$$\sum_{\substack{w \in W(\alpha) \\ w(n)=1}} \binom{\mathrm{ceq}(w)}{k - \mathrm{cdes}(w)}. \tag{21}$$

Proof. Let $F \in \mathcal{T}_{k,n}$ and w_F be as in Lemma 2.7. By (1), $\mathrm{color}(I_{i+1}) = \mathrm{color}(I_i) + |A_i|$. Therefore, $\alpha(\mathrm{type}(F)^{\mathrm{color}(\mathrm{base}(F))}) = (|A_1|, \ldots, |A_p|)$. By (17), w_F is the reading word of $(A_1, \ldots, A_p)$, so $\mathrm{content}(w_F) = \alpha(\mathrm{type}(F)^{\mathrm{color}(\mathrm{base}(F))})$. The definition of $\mathrm{base}(F)$ forces $w_F(n) = p$. For each such word w, Lemma 2.7 (4) shows that there are $\binom{\mathrm{ceq}(w)}{k-\mathrm{cdes}(w)}$ possible choices of $\mathrm{base}(F)$. □

Corollary 2.10. *Let* $[\alpha]$ *be a cyclic composition. Then*

$$\sum_{\substack{S \subseteq [n] \\ \mathrm{ccomp}(S) \in [\alpha]}} f_S(\mathcal{T}_{k,n}) = \sum_{\beta \in [\alpha]} \sum_{\substack{w \in W(\beta) \\ w(n)=1}} \binom{\mathrm{ceq}(w)}{k - \mathrm{cdes}(w)}. \tag{22}$$

Proof. Summing over all $\mathrm{comp}(F) = \beta \in [\alpha]$ in Proposition 2.9, we obtain the desired equality. □

Theorem 2.11. *For all* $0 < k < n$ *and every nonempty subset* $S \subseteq [n]$ *with* $|S| = p$, *over* $\mathbb{Q}[q]/(q^n - 1)$, *we have*

$$f_S(\mathcal{T}_{k,n}) = \sum_{i \in S} [q^i] \sum_{\substack{\mathrm{content}(w) = \mathrm{comp}^{(i)}(S) \\ w(n)=1}} q^{\mathrm{cmaj}(w)} e_{k-\mathrm{des}(w)} \left(q^j : j \in \mathrm{cEq}(w)\right) \tag{23}$$

where

$$\mathrm{cmaj}(w) := \sum_{i \in \mathrm{cDes}(w)} i \tag{24}$$

is the cyclic major index

Proof. Let F be a face with $\mathrm{type}(F) = S$, and let w_F be as in Lemma 2.7. Since $\mathrm{type}(F) = S$, there exists $i \in S$ such that $\mathrm{color}(\mathrm{base}(F)) = i$. By Lemma 2.7, $\mathrm{cDes}(w_F) \subseteq \mathrm{base}(F) \subseteq \mathrm{cDes}(w_F) \cup \mathrm{cEq}(w_F)$, so the inner sum is a generating function of $q^{\sum_{j \in \mathrm{base}(F)} j}$. □

Lemma 2.12. *Let $\alpha = (\alpha_1, \ldots, \alpha_p) \models n$. Then*

$$\alpha_1 \sum_{w \in W(\alpha)} \binom{\mathrm{ceq}(w)}{k - \mathrm{cdes}(w)} = n \sum_{\substack{w \in W(\alpha) \\ w(n)=1}} \binom{\mathrm{ceq}(w)}{k - \mathrm{cdes}(w)}. \tag{25}$$

Proof. The $\mathbb{Z}/n\mathbb{Z}$ action on cyclic words generated by

$$w \mapsto w', \qquad w'(i) = w(i-1), \tag{26}$$

preserves cdes and ceq. There are α_1 many i such that $w(i) = 1$, so the left hand side counts the number of pairs (w, i) for which $w \in W(\alpha)$ and $w(i) = 1$, weighted by the binomial coefficient. The right hand side applies the $\mathbb{Z}/n\mathbb{Z}$ action to w with $w(n) = 1$ and counts the same set of pairs with the same weight. □

3. Symmetric function identities

Our main method of proving the main theorem is via symmetric function identities. We draw inspirations from Ellzey and Wachs's quasisymmetric functions counting cyclic descents on Smirnov words [Ell17, EW20]. Let

$$G_{k,n} = \sum_{w : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}_{>0}} \binom{\mathrm{ceq}(w)}{k - \mathrm{cdes}(w)} \prod_{i \in \mathbb{Z}/n\mathbb{Z}} x_{w(i)} \tag{27}$$

be a power series in $x_1, x_2, \ldots$.

Let

$$\mathrm{Hyp}_{k,n} = \sum_{\substack{\alpha \in \mathbb{Z}_{\geq 0}^{\mathbb{Z}/k\mathbb{Z}}, \\ \alpha \models n, \\ \alpha \text{ is hypersimplicial}}} h_\alpha \tag{28}$$

By Definition 1.10, we have

$$\mathrm{Hyp}_{k,n}(x_1, \ldots, x_n) = \sum_{\mathsf{S} \in \mathrm{hDOSP}(k,n)} h_{s_1} \cdots h_{s_p} \tag{29}$$

where the second sum is over all $\mathsf{S} \in \mathrm{hDOSP}(k, n)$ with decorations $(s_1, \ldots, s_p) \models k$.

Let Θ be the $n \times n$ -matrix with entries

$$\Theta_{ij} = \begin{cases} 1, & i < j, \\ 1 + z, & i = j, \\ z, & i > j, \end{cases} \tag{30}$$

and let X be the diagonal matrix with $x_1, \ldots, x_n$ along the diagonal.

Theorem 3.1. *For all $n \geq 2$ and $1 \leq k \leq n-1$, we have*

$$G_{k,n} = \frac{n}{k} \mathrm{Hyp}_{k,n} \tag{31}$$

Lemma 3.2.

$$\sum_{k \geq 0} G_{k,n}(x_1, \ldots, x_n) z^k = \mathrm{tr}((\Theta X)^n). \tag{32}$$

Proof. Expanding the trace of the n -th power of ΘX, the trace is

$$\mathrm{tr}((\Theta X)^n) = \sum_{w : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}_{>0}} \prod_{i \in \mathbb{Z}/n\mathbb{Z}} \Theta_{w(i)w(i+1)} x_{w(i+1)} \tag{33}$$

the sum being over all cyclic words of length n. If $w(i) < w(i+1)$, then $\Theta_{w(i)w(i+1)} = 1$; if $w(i) = w(i+1)$, i.e. $i \in \mathrm{cEq}(w)$, then $\Theta_{w(i)w(i+1)} = 1+z$; if $w(i) > w(i+1)$, i.e. $i \in \mathrm{cDes}(w)$, then $\Theta_{w(i)w(i+1)} = z$. Therefore,

$$\mathrm{tr}((\Theta X)^n) = \sum_{w:\mathbb{Z}/n\mathbb{Z}\to\mathbb{Z}} z^{\mathrm{cdes}(w)}(1+z)^{\mathrm{ceq}(w)}x^w = \sum_{w:\mathbb{Z}/n\mathbb{Z}\to\mathbb{Z}}\sum_{k\geq 0}\binom{\mathrm{ceq}(w)}{k-\mathrm{cdes}(w)}z^k x^w. \tag{34}$$

□

Let

$$H(t) = \prod_{i\geq 1}\frac{1}{1-x_i t} = \sum_{n\geq 0} h_n t^n. \tag{35}$$

Then

$$\frac{1}{H(t)} = \prod_{i\geq 1}(1-x_i t) = \sum_{j\geq 0}(-1)^j e_j t^j. \tag{36}$$

Lemma 3.3. *Then*

$$\det(I - t\Theta X) = \frac{1}{1-z}\left(\frac{1}{H(t)} - \frac{z}{H(zt)}\right). \tag{37}$$

Proof. Let $\Theta^{(j)}$ be the principal $j \times j$ -minor of Θ. The determinants of $\Theta^{(j)}$ satisfy the following recursion.

$$\det\Theta^{(j)} = (1+z)\det\Theta^{(j-1)} - z\det\Theta^{(j-2)} \tag{38}$$

with initial conditions $\det\Theta^{(0)} = 1$ and $\det\Theta^{(1)} = 1+z$. By induction, we can show that $\det\Theta^{(j)} = [j+1]_z = 1 + z + \cdots + z^j = \frac{1-z^{j+1}}{1-z}$. We expand the determinant

$$\det(I - t\Theta X) = \sum_{j\geq 0}(-t)^j \sum_{|S|=j}\det(\Theta_{S,S})x^S. \tag{39}$$

Then

$$\begin{aligned}\det(I - t\Theta X) &= \sum_{j\geq 0}(-1)^j [j+1]_z e_j t^j \\ &= \frac{1}{1-z}\left(\sum_{j\geq 0}(-1)^j e_j t^j - z\sum_{j\geq 0}(-1)^j e_j (zt)^j\right) \\ &= \frac{1}{1-z}\left(\frac{1}{H(t)} - \frac{z}{H(zt)}\right).\end{aligned} \tag{40}$$

□

Lemma 3.4. *We have*

$$\sum_{n\geq 1}\mathrm{Hyp}_{k,n}(x_1,\ldots,x_n)t^n = k[z^k]\log\frac{1-z}{H(zt) - zH(t)}. \tag{41}$$

Proof. Compute the right hand side by

$$\begin{aligned}H(zt) - zH(t) &= \sum_{n\geq 0} h_n (zt)^n - z\sum_{n\geq 0} h_n t^n \\ &= h_0(1-z) + z\sum_{n\geq 2}(z^{n-1}-1)h_n t^n = (1-z) - z(1-z)\sum_{n\geq 0}[n-1]_z h_n t^n.\end{aligned} \tag{42}$$

Therefore,

$$\log\frac{1-z}{H(zt) - zH(t)} = \sum_{p\geq 1}\frac{1}{p}\left(z\sum_{n\geq 0}[n-1]_z h_n t^n\right)^p. \tag{43}$$

Observe that

$$z\sum_{n\geq 0}[n-1]_z h_n t^n = \sum_{j\geq 1} z^j \sum_{n\geq j+1} h_n t^n. \tag{44}$$

The coefficient of z^k in $\left(\sum_{j\geq 1} z^j \sum_{n\geq j+1} h_n t^n\right)^p$ is

$$\sum_{\mu}\sum_{\lambda} h_\lambda t^{|\lambda|} \tag{45}$$

where $\mu = (\mu_1, \ldots, \mu_p)$ is a composition of k, $\lambda = (\lambda_1, \ldots, \lambda_p)$ is composition such that $\lambda_i \geq \mu_i + 1$ for every $i \in \mathbb{Z}/p\mathbb{Z}$. For any hypersimplicial weak composition α, let λ be the composition consisting of only positive parts in α in order, and let μ be the composition such that μ_i is the circular distance between λ_i and λ_{i+1}. Then (45) is equal to

$$\sum_{\substack{\alpha\in\mathbb{Z}_{\geq 0}^{\mathbb{Z}/k\mathbb{Z}},\\ \alpha\models n,\\ \alpha \text{ is hypersimplicial}}} h_\alpha$$

which is equal to the left hand side. □

Proof of Theorem 3.1. By Lemma 3.2

$$-\log\det(I - t\Theta X) = \sum_{n\geq 1} \frac{\operatorname{tr}((\Theta X)^n)t^n}{n} = \sum_{n\geq 1}\sum_{k\geq 0} \frac{G_{k,n} z^k t^n}{n} \tag{46}$$

By Lemma 3.3,

$$-\log\det(I - t\Theta X) = \log H(t) + \log H(zt) + \log\frac{1-z}{H(zt) - zH(t)} \tag{47}$$

$$= \log H(t) + \log H(zt) + \sum_{n\geq 1}\sum_{k\geq 0} \frac{\operatorname{Hyp}_{k,n}\, t^n z^k}{k}. \tag{48}$$

The error terms $\log H(t)$ and $\log H(zt)$ only contribute to the coefficients of $z^k t^n$ with $k = 0$ or $k = n$, so the equality holds for all $1 \leq k \leq n-1$. □

Corollary 3.5. *The sum*

$$\sum_{w\in W(\alpha)} \binom{\operatorname{ceq}(w)}{k - \operatorname{cdes}(w)} = \langle h_\alpha, G_{k,n}\rangle$$

depends only on the multiset of parts of α.

Proof. By definition, $\operatorname{Hyp}_{k,n}$ is a symmetric function. By Theorem 3.1, $G_{k,n}$ is also a symmetric function. The definition of $G_{k,n}$ expands it into the monomial basis $\{m_\alpha\}$ of the ring of symmetric functions. Since the monomial symmetric function is dual to the complete homogeneous symmetric function, taking inner product with h_α extracts the coefficient of m_α, which is the left hand side. □

Theorem 3.6. *For all $0 < k < n$ and every nonempty subset $S \subseteq [n]$ with $\operatorname{ccomp}(S) = [\alpha]$, we have*

$$\sum_{j\in\mathbb{Z}/n\mathbb{Z}} f_{S+j}(\mathcal{T}_{k,n}) = \langle h_\alpha, G_{k,n}\rangle. \tag{49}$$

Proof. Let

$$N(\alpha) = \sum_{\substack{w\in W(\alpha)\\ w(n)=1}} \binom{\operatorname{ceq}(w)}{k - \operatorname{cdes}(w)}.$$

By Lemma 2.12,

$$\sum_{j\in\mathbb{Z}/n\mathbb{Z}} f_{S+j}(\mathcal{T}_{k,n}) = \sum_{i=0}^{p-1} N(\alpha^{(i)}) \tag{50}$$

$$= \frac{1}{n}\sum_{i=0}^{p-1} \alpha_i \sum_{w\in W(\alpha^{(i)})} \binom{\operatorname{ceq}(w)}{k - \operatorname{cdes}(w)} = \sum_{w\in W(\alpha)} \binom{\operatorname{ceq}(w)}{k - \operatorname{cdes}(w)}$$

because $\langle h_{\alpha^{(i)}}, G_{k,n}\rangle = \langle h_\alpha, G_{k,n}\rangle$ for any $i \in \mathbb{Z}/p\mathbb{Z}$ and $\sum \alpha_i = n$. □

Proof of Theorem 1.12. If $\gcd(k,n) = 1$, by Corollary 2.6, the left hand side of Theorem 3.6 is equal to $nf_S(\mathcal{T}_{k,n})$. By Theorem 3.1, the right hand side is equal to $\frac{n}{k}\langle h_\alpha, \operatorname{Hyp}_{k,n}\rangle$. The character of M^α is h_α, and the character of $\mathcal{M}_{k,n}$ is $\operatorname{Hyp}_{k,n}$, so the dimension of $\operatorname{Hom}_{S_n}(M^\alpha, \mathcal{M}_{k,n})$ is equal to $\langle h_\alpha, \operatorname{Hyp}_{k,n}\rangle$. □

By Corollary 2.6, we have the following fact.

Corollary 3.7. *Let $d = \gcd(k, n)$, and let S be a nonempty subset of $[n]$ with $\mathrm{ccomp}(S) = [\alpha]$. Then $f_{S+j}(\mathcal{T}_{k,n})$ depends on j only through $j \bmod d$, and*

$$\frac{n}{d}\sum_{j=0}^{d-1} f_{S+j}(\mathcal{T}_{k,n}) = \sum_{w\in W(\alpha)} \binom{\mathrm{ceq}(w)}{k-\mathrm{cdes}(w)}. \tag{51}$$

4. Hypersimplicial words

Theorem 4.1. *Let $S \subseteq [n]$ and $\mathrm{comp}(S) = \alpha$. Let W_S be the set of hypersimplicial words w with $\mathrm{cDes}(w) \subseteq S$. The cardinality of W_S is*

$$|W_S| = \langle h_\alpha, \mathrm{Hyp}_{k,n}\rangle. \tag{52}$$

Recall that $\langle h_\alpha, h_\lambda\rangle$ is equal to the number of matrices with nonnegative integer entries with row sum α and column sum λ. The set of matrices with fixed row sum vector and column sum vector are also called *contingency tables*.

Proof. We construct a bijection between hypersimplicial words w with $\mathrm{cDes}(w) \subseteq S$, content λ, and contingency tables with row sum α and column sum λ. Let (a_{ij}) be such a contingency table, with row sum α and column sum λ. Let w be the word with a_{11} ones followed by a_{12} twos, etc., up to a_{1n} n's, then a_{21} ones, a_{22} twos, etc. By construction, $\mathrm{cDes}(w) \subseteq S$ for any S with $\mathrm{comp}(S) = \alpha$ and $\mathrm{content}(w) = \lambda$. □

The p-shift (8) gives a bijection between $\mathrm{hDOSP}(k, n) \times \mathbb{Z}/k\mathbb{Z}$ onto hypersimplicial words via

$$(\mathsf{S}, p) \mapsto w_{\mathsf{S}}^{(p)}.$$

Since the winding number is a cyclic invariant, we have $|\,\mathrm{cDes}(w_{\mathsf{S}}^{(p)})| = \mathrm{wind}(\mathsf{S})$ for any $p \in \mathbb{Z}/k\mathbb{Z}$.

Proof of Theorem 1.11. By Möbius inversion,

$$f_S(\mathcal{T}_{k,n}) = \sum_{T\subseteq S} h_T(\mathcal{T}_{k,n}),$$

so it suffices to prove that

$$\sum_{S:\mathrm{ccomp}(S)=[\alpha]} f_S(\mathcal{T}_{k,n}) = \sum_{S:\mathrm{ccomp}(S)=[\alpha]} \left|\{\mathsf{S} \in \mathrm{hDOSP}(k, n) : \mathrm{cDes}(\mathsf{S}) \subseteq S\}\right|. \tag{53}$$

By the bijection from $\mathrm{hDOSP}(k, n) \times \mathbb{Z}/k\mathbb{Z}$ onto hypersimplicial words via $(\mathsf{S}, p) \mapsto w_{\mathsf{S}}^{(p)}$, we multiply both sides of (53) by k to obtain

$$\begin{aligned} k\sum_{S:\mathrm{ccomp}(S)=[\alpha]} f_S(\mathcal{T}_{k,n}) &= \sum_{p\in\mathbb{Z}/k\mathbb{Z}}\sum_{S:\mathrm{ccomp}(S)=[\alpha]} \left|\left\{\mathsf{S} \in \mathrm{hDOSP}(k, n) : \mathrm{cDes}(w_{\mathsf{S}}^{(p)}) \subseteq S\right\}\right| \\ &= \sum_{S:\mathrm{ccomp}(S)=[\alpha]} \left|\left\{(\mathsf{S}, p) \in \mathrm{hDOSP}(k, n) \times \mathbb{Z}/k\mathbb{Z} : \mathrm{cDes}(w_{\mathsf{S}}^{(p)}) \subseteq S\right\}\right| \\ &= \sum_{S:\mathrm{ccomp}(S)=[\alpha]} W_S. \end{aligned} \tag{54}$$

Fix some $S \subseteq [n]$ with $\mathrm{ccomp}(S) = [\alpha]$ and let $O(S) = \{S + j : j \in \mathbb{Z}/n\mathbb{Z}\}$ be the orbit of S under the $\mathbb{Z}/n\mathbb{Z}$ action, and let $\mathrm{Stab}(S)$ be the stabilizer of S under this action. By the orbit-stabilizer theorem,

$$\sum_{j\in\mathbb{Z}/n\mathbb{Z}} f_{S+j}(\mathcal{T}_{k,n}) = |\,\mathrm{Stab}(S)| \sum_{S'\in O(S)} f_{S'}.$$

The set $\{S \subseteq [n] : \mathrm{ccomp}(S) = [\alpha]\}$ is a union of such orbits, so it suffices to show that

$$\sum_{S'\in O(S)} f_{S'} = \sum_{S'\in O(S)} |W_{S'}|.$$

Again by the orbit stabilizer theorem,

$$|\,\mathrm{Stab}(S)| \sum_{S'\in O(S)} |W_{S'}| = \sum_{j\in\mathbb{Z}/n\mathbb{Z}} |W_{S+j}|.$$

However, if $\mathrm{ccomp}(S) = [\alpha]$, then $\mathrm{ccomp}(S + j) = [\alpha]$, so the right hand side is equal to $n|W_S| = n\langle h_\alpha, \mathrm{Hyp}_{k,n}\rangle$ by Theorem 4.1. By Theorem 3.6 we reduce the theorem to

$$k\langle h_\alpha, G_{k,n}\rangle = n\langle h_\alpha, \mathrm{Hyp}_{k,n}\rangle$$

which holds by Theorem 3.1. □

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